User manual

SparkFlow

SparkFlow bridges the Sparkplug B stream Host decodes onto a NATS/JetStream backbone, so every point stays live and available company-wide, with no tag behind it, past the point where a tag-counted system stops scaling.

Version 3.0.0Ignition 8.1.19+ and 8.3Runs on: the Host gateway
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Overview

1. Overview

SparkFlow publishes every point Host decodes onto a NATS JetStream stream as a named, timestamped, quality-stamped message, independent of whether a tag exists for it. Dashboards, historians, analysts and AI pipelines subscribe directly; the plant is asked for nothing extra. A key-value store holds the current value and quality of every point, so a new subscriber gets the last known value of everything the moment it connects, and the stream keeps history so a window of past data can be replayed. SparkFlow also carries a command bridge: a NATS request/reply call can drive a confirmed NCMD/DCMD write back to the field.

SparkFlow is optional and is not part of the core SparkBridge install. It runs on the Host gateway, on its own perpetual per-gateway license, and is installed after Provider, Host and Edge are running.

Requirements

2. What You Need

Ignition	8.1.19 or newer on the 8.1 line, or 8.3 (separate build). A gateway that can be restarted at will.
Java	Compiles against the Ignition 8.1.0 API, targets Java 11 bytecode.
Gateway role	The Host gateway. SparkFlow is not installed on an Edge-only gateway.
Network	A reachable `nats-server` with JetStream enabled, default `nats://127.0.0.1:4222`.
Other SparkBridge modules required	Provider, Host and Edge running first. SparkFlow reads from Host's decoded stream.

Install

3. Install

1. Download the release bundle for your Ignition line from the download page, including `SparkBridge-Flow.modl` (optional add-on, bought separately from the core bundle).
2. Install after Provider, Host and Edge are running: Config > Modules > Install or Upgrade a Module, upload `SparkBridge-Flow.modl`.
3. Accept the certificate and license once; confirm Config > Modules shows SparkFlow state = Running.

See `docs/FLOW_MODULE.md` in the bundle for the full configuration and lifecycle reference.

License

4. License

File	Per-gateway license key at `<data>/sparkbridge-license.key`. SparkFlow carries its own license, separate from the core suite key, at $995 per Host gateway.
Re-check interval	SparkFlow re-checks the key file every 30 seconds. Installing, replacing or removing a key takes effect with no restart.

From release 2.7.0, SparkFlow runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). When the trial expires with no verified key naming the product, SparkFlow stops its dispatcher. A key dropped in after expiry lifts the pause on its own within the 30-second re-check, no button press and no restart. A verified key overrides the trial entirely.

Configure

5. Configure

All keys are `-D` system properties, read at startup as bootstrap/defaults. The `tagfeed` filter additionally hot-reloads from the `sparkflow-config` NATS KV bucket at runtime; nothing else currently hot-reloads.

Setting	Default	Meaning
`sparkbridge.flow.enabled`	true	Master on/off switch. When false, startup logs and returns without connecting to NATS or the broker.
`sparkbridge.flow.nats.url`	`nats://127.0.0.1:4222`	NATS core connect URL.
`sparkbridge.flow.tagfeed`	empty	Comma-separated NATS-subject-style filter list (`*` = one token, `>` = remainder) selecting the subset projected onto the managed tag provider. Empty means nothing projects. Hot-reloads from the `sparkflow-config` KV bucket.
`sparkbridge.flow.allowWildcardTagfeed`	false	Must be explicitly set to true before a tagfeed filter list may contain the whole-estate wildcard `uns.>`.
`sparkbridge.flow.subjectprefix`	`uns.`	Telemetry subject prefix. Present for future configurability; the shipped

Setting	Default	Meaning
		subject mapper currently hardcodes `uns.`, so changing this key alone has no effect on the wire format.
`sparkbridge.flow.kvbucket`	`uns-last`	KV bucket holding tag-less last-known-value and lifecycle state. Created automatically if absent.
`sparkbridge.flow.configbucket`	`sparkflow-config`	KV bucket SparkFlow watches for its own hot-reloadable config. Not created automatically; must be provisioned ahead of time. If absent at startup, SparkFlow logs an error and continues on the static bootstrap tagfeed filter with hot-reload disabled.
`sparkbridge.flow.twin.kvbucket`	`uns-model`	Digital-twin KV bucket. Created automatically if absent, unlike the config bucket.
`sparkbridge.flow.commandTimeoutMs`	5000	Timeout for the reverse (NATS request/reply to NCMD/DCMD) command path.
`sparkbridge.flow.commandWorkers`	8	Threads in the command-bridge worker pool.
`sparkbridge.flow.commandQueue`	64	Bounded queue in front of the command workers. When full, a new command is answered at once with `{"busy":true}` rather than queued.
`sparkbridge.flow.commandPerEdge`	2	Commands allowed in flight per group/edge. A dead edge pins at most this many workers.
`sparkbridge.flow.nats.credentialsFile`	none	NATS credentials file path, resolved through the secrets file in preference to a bare `-D` value.
`sparkbridge.flow.nats.tls.enabled`	false	Whether the NATS connection uses TLS.
`sparkbridge.flow.nats.tls.certPath` / `keyPath` / `caPath`	none	TLS cert/key/CA paths, resolved through the secrets file the same way as the credentials file.
`sparkbridge.flow.cmd.keysDir`	empty	Directory of signed-command verification keys (public keys only, not routed through the secrets file).
`sparkbridge.flow.cmd.requireSignature`	false	When true, an unsigned command request is rejected instead of falling back to the unsigned path.

Every subject, command subject and KV key includes a mandatory device slot: uns.<group>.<edge>.<device|_>.<metric segment>[...], with a literal underscore meaning no device. Metric segment characters outside `[A-Za-z0-9_-]` are percent-style escaped as `=XX` (two hex digits); a literal `=` becomes `=3D`. This escaped form is canonical on the wire subject, the KV key, and the historian node key.

Operate

6. Operate

Confirm the Edge's Sparkplug traffic reaches NATS subjects under the `uns.` prefix, for example with a NATS CLI subscription to `uns.>`. Watch for repeated `{"busy":true}` command responses, which indicate the command queue or per-edge limit is saturated rather than a dead command bridge.

On a NATS link loss, SparkFlow's dispatcher stops publishing until reconnected; it does not buffer independently of what NATS/JetStream itself retains. On restart, SparkFlow reconnects to NATS and re-subscribes to Host's decoded stream; the tagfeed filter reloads from the config bucket if present.

A metric segment that is permanently unmappable for an unsupported wire value type or a rejected group/edge/device identifier publishes one event to `uns.$events.reject` and is never queued for retry.

Verify

7. Verify

Confirm the Edge's Sparkplug traffic reaches NATS subjects under the `uns.` prefix, for example with a NATS CLI subscription to `uns.>`. See `docs/FLOW_MODULE.md` in the bundle for the subject naming contract and the KV bucket layout used for last-known values.

Troubleshoot

8. Troubleshoot

Symptom	Likely cause	Fix
No subjects appear under `uns.`	`sparkbridge.flow.enabled` is false, or `tagfeed` is empty	Confirm the module is enabled and set a non-empty tagfeed filter
Startup logs an error naming the config bucket	`sparkflow-config` KV bucket was never provisioned	Provision the bucket ahead of time; SparkFlow will not create it itself
Command calls return `{"busy":true}` repeatedly	Command queue full or per-edge limit reached on a dead/slow edge	Investigate the named edge; raise `commandWorkers/commandQueue` only after confirming the edge itself is healthy
Metric with spaces or punctuation in its name fails to publish	Pre-escaping defect in an old build, or an unsupported wire value type	Confirm the module escapes metric segments as `=XX`; an unsupported value type instead routes to `uns.$events.reject`
Dispatcher stops publishing with no error	Module trial expired, no verified key	Drop a valid `sparkbridge-license.key` in the data directory

Reference

9. Reference

License file	`<data>/sparkbridge-license.key`
Module file	`SparkBridge-Flow.modl`
Default NATS URL	`nats://127.0.0.1:4222`
Last-value KV bucket	`uns-last`
Config KV bucket	`sparkflow-config` (not auto-created)
Digital-twin KV bucket	`uns-model`

Reject event subject	`uns.$events.reject`
License re-check	30 seconds

Support

10. Support

Full install and config detail: SparkBridge Docs, Flow. Product page: SparkFlow. Questions: Contact. Release history: GitHub releases.